Unit: Unit 8: Quadratics
Topic: Systems of Quadratic Equations

Name: ________________
Date: ____________

Advanced (Habanero)

Q1. Solve the system: $x^2 + y^2 = 4$, $x^2 - y^2 = -1$
(A) $(-1, -sqrt2)$
(B) $(-1, sqrt2)$
(C) $(1, -sqrt2)$
(D) $(1, sqrt2)$
(E) No solution

Q2. Solve the system: $y = x^2 - 2$, $y = -x^2 + 4$
(A) $(-1, 1)$
(B) $(-1, 3)$
(C) $(1, 1)$
(D) $(1, 3)$
(E) $(-1, -1)$

Q3. Solve the system: $x^2 + y^2 = 25$, $x - y = 3$
(A) $(-2, -5)$
(B) $(-2, 5)$
(C) $(2, -5)$
(D) $(2, 5)$
(E) No solution

Q4. Graph the system: $y = x^2 - 4$, $x^2 + y^2 = 16$
(A) 1 intersection point
(B) 2 intersection points
(C) 3 intersection points
(D) 4 intersection points
(E) No intersection points

Q5. Solve the system: $y = x^2 - 2x - 3$, $y = x^2 + 2x - 3$
(A) $(-1, -4)$
(B) $(-1, -2)$
(C) $(1, -2)$
(D) $(1, 2)$
(E) $(-2, -1)$

Q6. Solve the system: $x^2 - y^2 = 3$, $x^2 + y^2 = 7$
(A) $(sqrt2, -sqrt5)$
(B) $(sqrt2, sqrt5)$
(C) $(-sqrt2, -sqrt5)$
(D) $(-sqrt2, sqrt5)$
(E) No solution

Q7. Graph the system: $y = x^2 - 3x - 2$, $y = -x^2 + 3x + 2$
(A) 1 intersection point
(B) 2 intersection points
(C) 3 intersection points
(D) 4 intersection points
(E) No intersection points

Q8. Solve the system: $x^2 + y^2 = 9$, $x - y = 1$
(A) $(-2, -1)$
(B) $(-2, 1)$
(C) $(2, -1)$
(D) $(2, 1)$
(E) No solution

Open Ended Questions (FRQ)

Q9. Solve the system $y = x^2 - 4x + 3$, $y = -x^2 + 4x - 1$ and graph the solutions.

Q10. Solve the system $x^2 + y^2 = 10$, $y = x^2 - 5$ and interpret the solutions graphically.

Chef's Tips

- Make sure to check for extraneous solutions when solving systems of quadratic equations.
- Use graphing to visualize the solutions and check for accuracy.
- Make sure students show their work and justify their answers when solving free response questions.